

Data Handling and Visualization

DSA060

Slot B

CO2 – AT1

Assessment Test 4

Chart Construction Test

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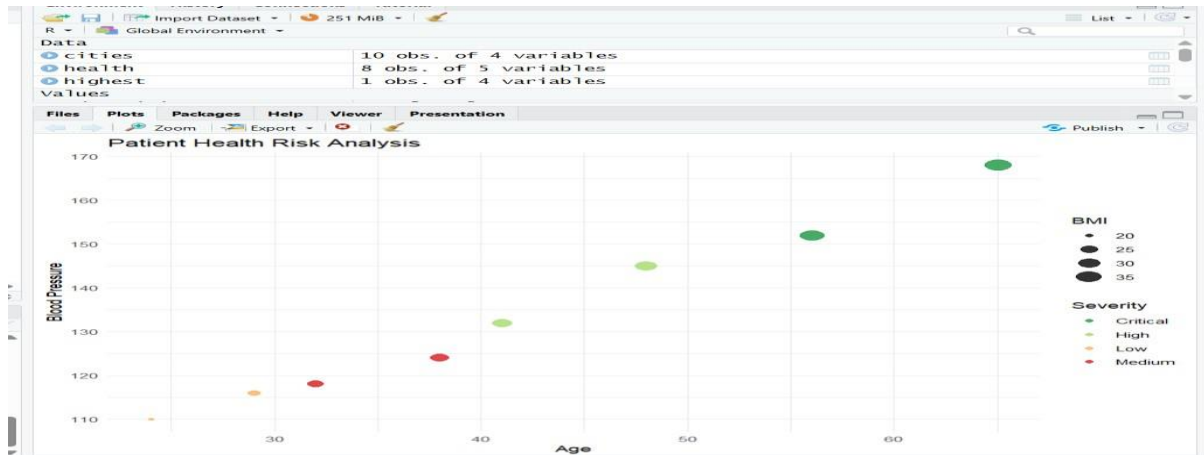
Question 1 (Healthcare Analytics)

```
# Load library
library(ggplot2)

# Dataset
health <- data.frame(
  Patient = c('P1', 'P2', 'P3', 'P4', 'P5', 'P6', 'P7', 'P8'),
  Age      = c(24, 32, 41, 48, 56, 65, 38, 29),
  BMI      = c(19.8, 24.5, 28.1, 30.3, 33.2, 36.5, 26.2, 22.1),
  BP       = c(110, 118, 132, 145, 152, 168, 124, 116),
  Severity = c('Low', 'Medium', 'High', 'High', 'Critical', 'Critical', 'Medium', 'Low')
)

# Bubble chart
ggplot(health, aes(x = Age, y = BP, color = Severity, size = BMI)) +
  geom_point(alpha = 0.8) +
  scale_color_brewer(palette = 'RdYlGn', direction = -1) +
  labs(
    title = 'Patient Health Risk Analysis',
    x      = 'Age',
    y      = 'Blood Pressure'
  ) +
  theme_minim
```

OutPut:



Justification

- Age → X-axis: shows increase in risk with age.
- Blood Pressure → Y-axis: primary clinical indicator.
- Severity → Color: red shades highlight critical patients quickly.
- BMI → Point size: larger circles indicate obesity-related risk.
- The chart simultaneously displays four health variables, making high-risk patients easy to identify.

Question 2 – Climate Science

Load library

```
library(ggplot2)
```

Dataset

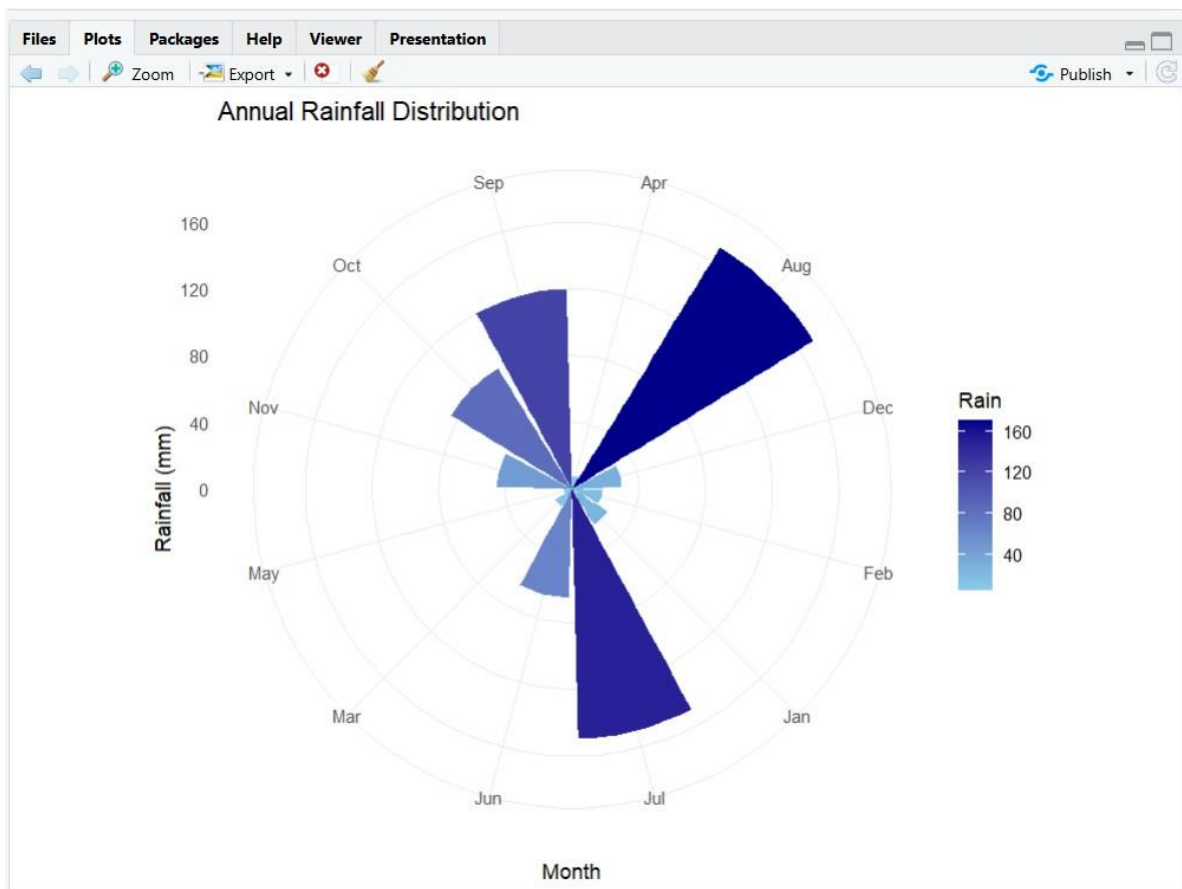
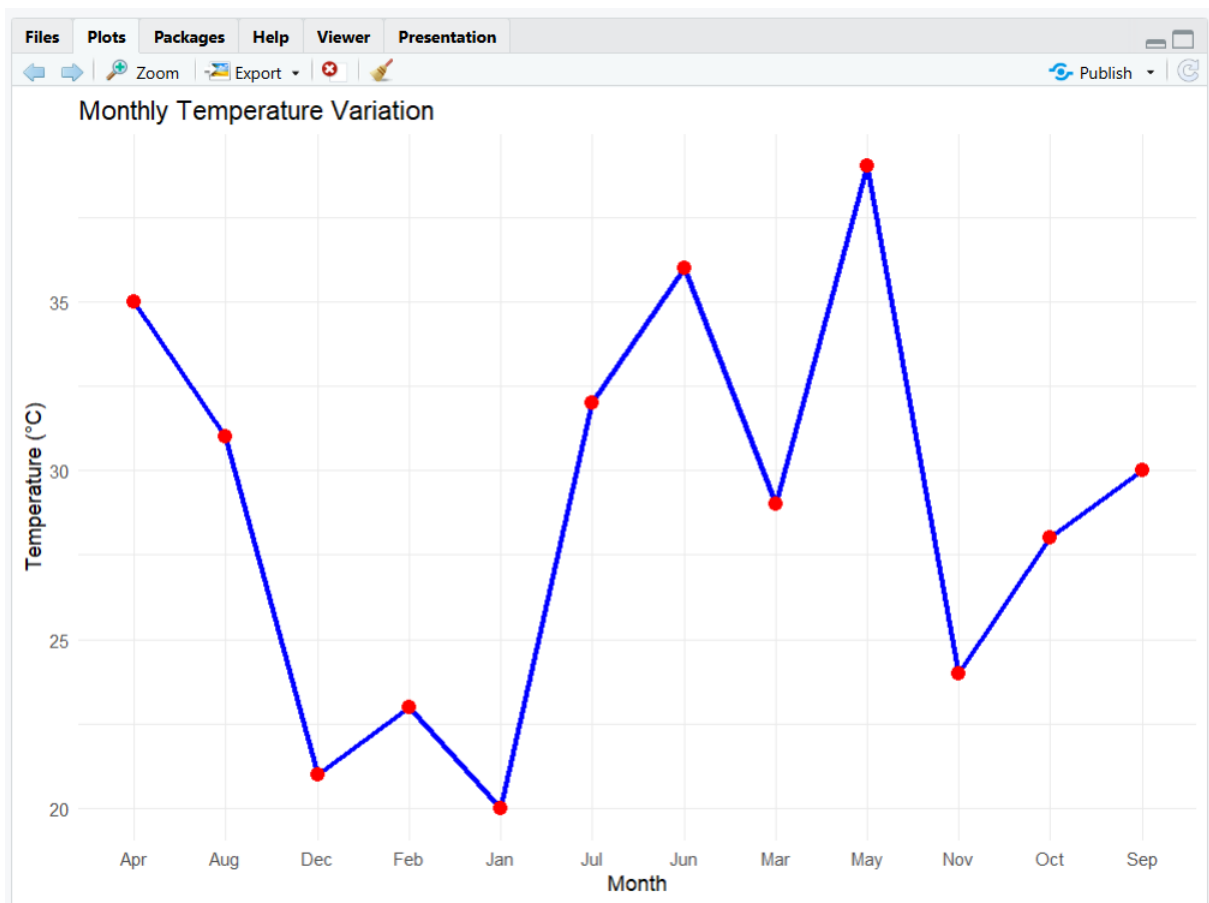
```
climate <- data.frame(  
  Month = c('Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'),  
  Solar = c(180, 200, 240, 280, 320, 300, 260, 250, 240, 220, 190, 170),  
  Temp = c(20, 23, 29, 35, 39, 36, 32, 31, 30, 28, 24, 21),  
  Rain = c(25, 18, 12, 8, 5, 65, 150, 170, 120, 85, 45, 30)  
)
```

1. Cartesian Line Chart

```
ggplot(climate, aes(x = Month, y = Temp, group = 1)) +  
  geom_line(linewidth = 1.2, color = 'blue') +  
  geom_point(size = 3, color = 'red') +  
  labs(  
    title = 'Monthly Temperature Variation',  
    x = 'Month',  
    y = 'Temperature (°C)'  
  ) +  
  theme_minimal()
```

2. Polar Coordinate Rainfall Chart

```
ggplot(climate, aes(x = Month, y = Rain, fill = Rain)) +  
  geom_bar(stat = 'identity') +  
  coord_polar() +  
  scale_fill_gradient(low = 'skyblue', high = 'darkblue') +  
  labs(  
    title = 'Annual Rainfall Distribution',  
    x = 'Month',  
    y = 'Rainfall (mm)'  
  ) +  
  theme_minimal()
```



Comparison

Cartesian chart advantages

- Accurate comparison of temperature values.
- Trends and seasonal changes are easy to interpret.

Cartesian limitations

- Does not emphasize cyclic annual patterns.

Polar chart advantages

- Clearly shows yearly seasonal rainfall cycle.
- Highlights monsoon months effectively.

Polar limitations

- Exact value comparison is less accurate than Cartesian charts.

Question 3 – Business Analytics & Marketing

```
# Load library
```

```
library(ggplot2)
```

```
# Dataset
```

```
business <- data.frame(
```

```
  Product = c('A', 'B', 'C', 'D', 'E'),
```

```
  Sales   = c(210, 185, 260, 145, 120),
```

```
  Profit  = c(18, 14, 25, 12, 10),
```

```
  Share   = c(24, 20, 28, 15, 13)
```

```
)
```

```
# Feature engineering
```

```
business$Highlight <- ifelse(business$Share == max(business$Share), 'Highest Share',  
'Other')
```

```
# Column chart with a conditional highlight overlay
```

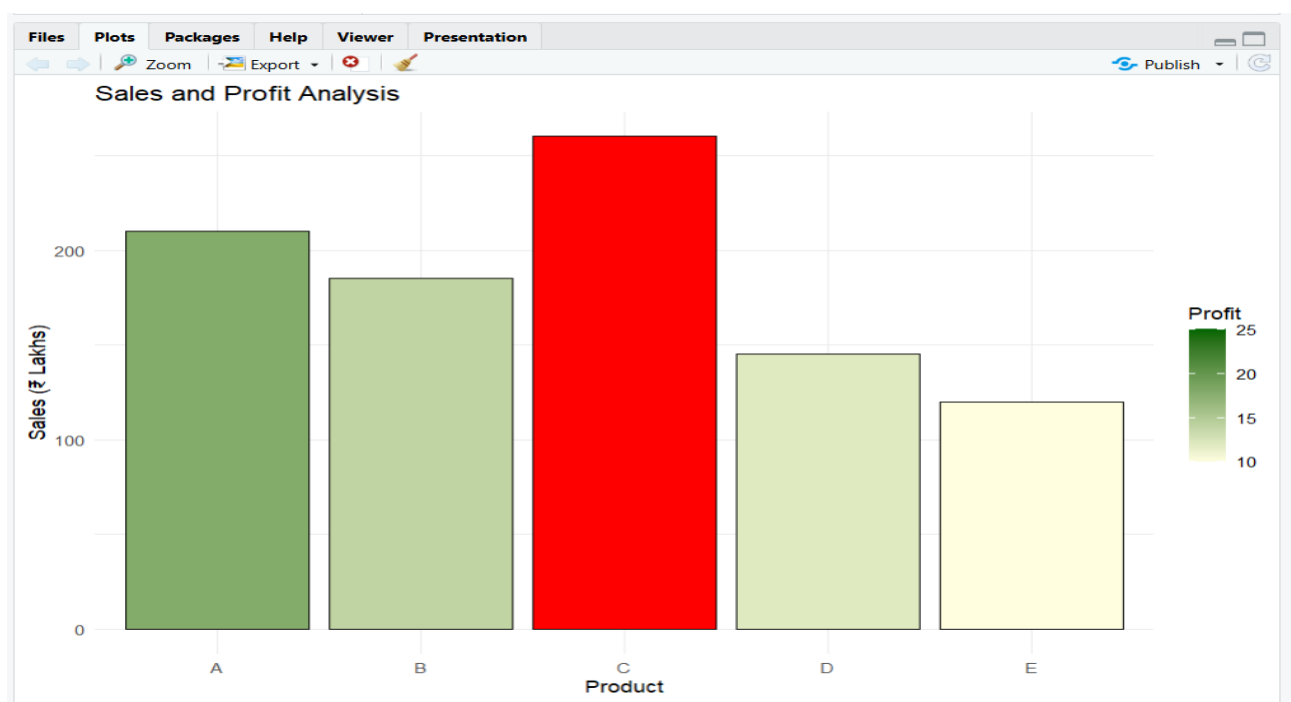
```
ggplot(business, aes(x = Product, y = Sales, fill = Profit)) +
```

```
geom_col(color = 'black') +
```

```

geom_col(data = subset(business, Highlight == 'Highest Share'), fill = 'red', color = 'black')
+
scale_fill_gradient(low = 'lightyellow', high = 'darkgreen') +
labs(
  title = 'Sales and Profit Analysis',
  x    = 'Product',
  y    = 'Sales (₹ Lakhs)'
) +
theme_minimal()

```



Interpretation

- Product C has the highest sales, highest profit, and highest market share.
- Sequential green shades indicate increasing profitability.
- Red highlight immediately draws management attention to the market leader.

Why this palette helps decision-making

- Darker green = better profit performance.
- Light colors indicate weaker products.
- Managers can prioritize high-profit products for investment.

Question 4 – Transportation Engineering

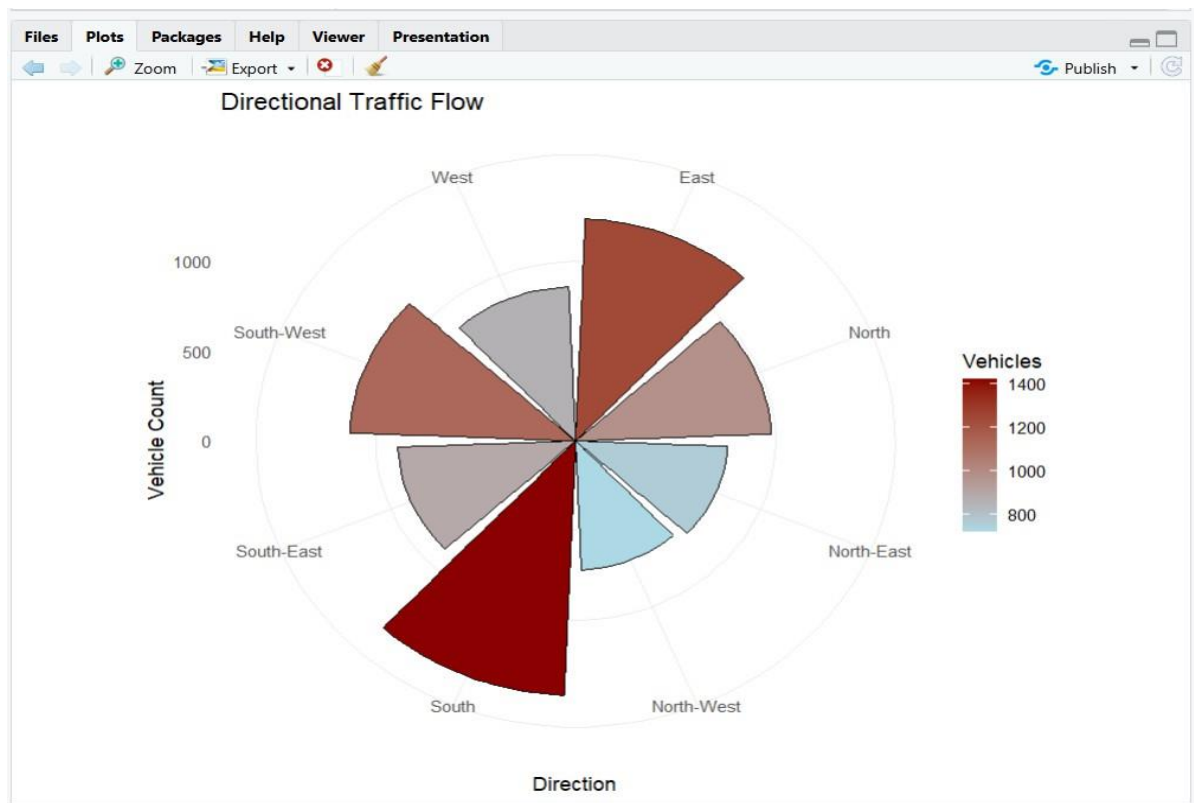
```

# Load library
library(ggplot2)

# Dataset
traffic <- data.frame(
  Direction = c('North', 'North-East', 'East', 'South-East', 'South', 'South-West', 'West', 'North-
West'),
  Vehicles = c(980, 760, 1240, 890, 1420, 1130, 860, 720)
)

# Wind rose style polar chart
ggplot(traffic, aes(x = Direction, y = Vehicles, fill = Vehicles)) +
  geom_bar(stat = 'identity', color = 'black') +
  coord_polar() +
  scale_fill_gradient(low = 'lightblue', high = 'darkred') +
  labs(
    title = 'Directional Traffic Flow',
    x = 'Direction',
    y = 'Vehicle Count'
  ) +
  theme_minimal()

```



Explanation

- Polar charts naturally represent **directional data** around a circle.
- Peak traffic directions become visually obvious.
- Darker colors indicate heavier traffic density.

Why Polar is better than Cartesian

- Compass directions are circular by nature.
- Spatial orientation is preserved.
- Direction-to-direction comparison is more intuitive.

Question 5 – Environmental Sustainability Dashboard

Load libraries

```
library(ggplot2)
```

```
library(gridExtra)
```

Dataset

```
env <- data.frame(
  City = c('A', 'B', 'C', 'D', 'E', 'F'),
```

```
AQI = c(48, 82, 135, 165, 72, 95),  
Green = c(45, 32, 21, 18, 38, 30),  
CO2 = c(1.8, 3.5, 5.6, 7.4, 2.9, 4.2),  
Pop = c(9, 15, 22, 28, 12, 18)  
)
```

```
# Feature engineering: AQI category mapping  
env$AQI_Category <- cut(  
  env$AQI,  
  breaks = c(0, 50, 100, 150, 200),  
  labels = c('Good', 'Moderate', 'Unhealthy', 'Very Unhealthy')  
)
```

```
# Plot 1: AQI by City  
p1 <- ggplot(env, aes(x = City, y = AQI, fill = AQI_Category)) +  
  geom_col() +  
  scale_fill_manual(values = c('green', 'yellow', 'orange', 'red')) +  
  labs(  
    title = 'AQI by City',  
    x = 'City',  
    y = 'Air Quality Index'  
  ) +  
  theme_minimal()
```

```
# Plot 2: Green Cover vs CO2 Emission  
p2 <- ggplot(env, aes(x = Green, y = CO2, size = Pop, color = AQI)) +  
  geom_point(alpha = 0.8) +  
  scale_color_gradient(low = 'green', high = 'red') +  
  labs(  
    title = 'Green Cover vs CO2 Emission',
```

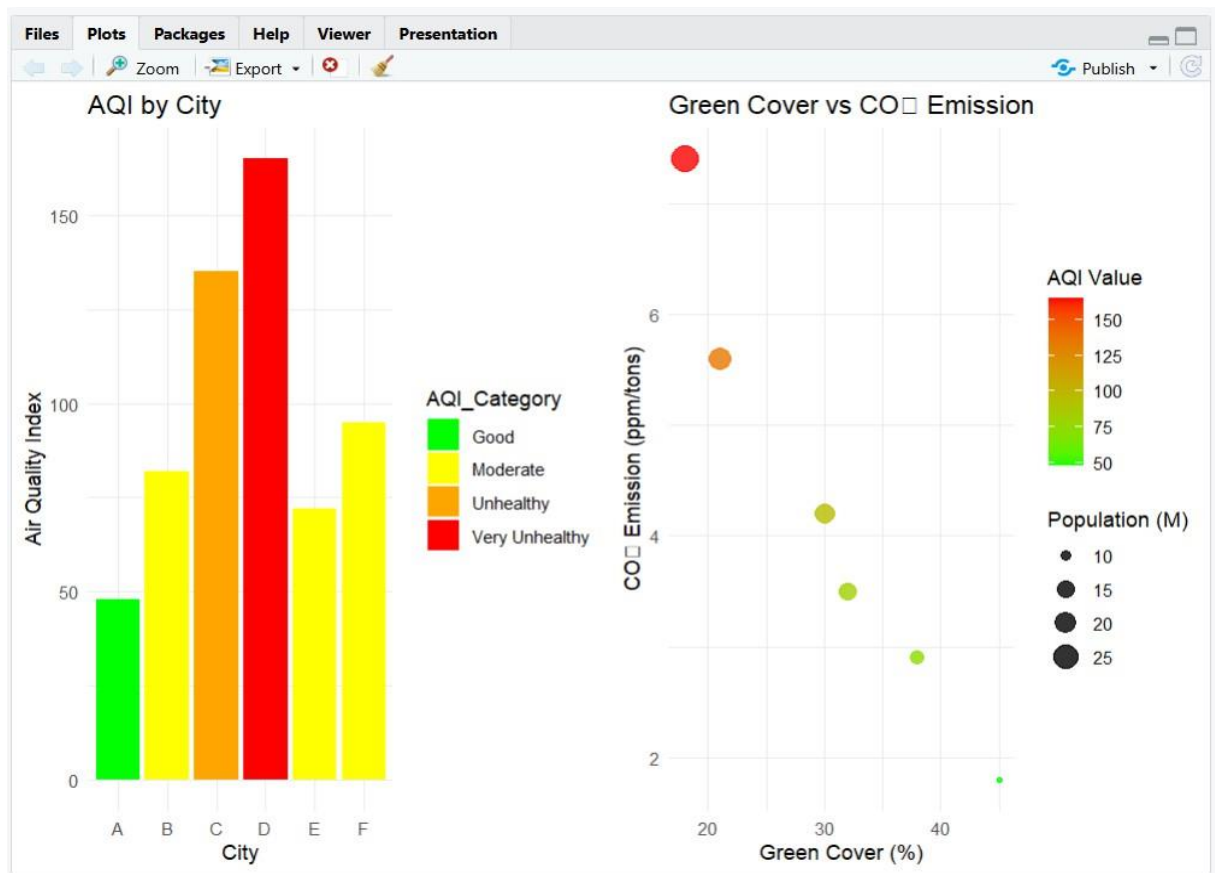
```

x = 'Green Cover (%)',
y = 'CO2 Emission (ppm/tons)',
size = 'Population (M)',
color = 'AQI Value'
) +
theme_minimal()

```

Arrange multiple plots in a grid layout

```
grid.arrange(p1, p2, ncol = 2)
```



Dashboard Design Justification

- Bar chart: compares AQI across cities.
- Scatter plot: shows relationship between green cover and CO₂ emissions.
- Point size: represents population impact.
- AQI color palette: green → yellow → orange → red follows standard air-quality interpretation.
- Continuous gradient: higher AQI appears more alarming.

Interpretation

- City D has the worst air quality and highest CO₂ emission.
- Higher green cover generally corresponds to lower CO₂ emission.
- Population helps identify cities with larger environmental impact.

Conclusion

These visualizations use appropriate axes, color palettes, scales, and chart types to communicate healthcare risk, climate patterns, business performance, traffic flow, and environmental sustainability effectively while integrating domain knowledge with R programming and data visualization.